Rajat Baranwal

Software Engineer — Full Stack Developer — Backend Engineering — AI Integration
+91-9934275919 | rajat123baranwal@gmail.com | LinkedIn | GitHub | Portfolio

Education

SRM University, Andhra Pradesh 2022 – 2026
Bachelor of Technology in Computer Science and Engineering | CGPA: 8.24/10 *Merit Scholarship: 85%*

Professional Experience

XiPHI AI — Seattle, WA, USA April 2026 – October 2026
Software Engineer Intern *Remote*

- Developed onboard.myeventz.ai, an attendee onboarding platform, implementing registration workflows and integrating FastAPI backend endpoints and n8n webhooks for attendee data submission and processing.
- Integrated Cloudflare Turnstile CAPTCHA verification and resolved production issues by troubleshooting Next.js environment variables, Docker build arguments, and container configurations.
- Developed frontend interfaces for real-time transcription and translation features within myEventz.ai, integrating FastAPI backend services to display live transcripts and translated content.
- Engineered real-time speech-to-text pipelines using Python, FastAPI, WebSockets, and ElevenLabs Scribe v2 Realtime, implementing asynchronous audio streaming and processing partial and finalized transcription events.
- Implemented PostgreSQL-backed translation caching to retrieve previously generated translations and avoid redundant Gemini-powered n8n workflow executions.
- Conducted API load and performance testing using Apache JMeter and Grafana to identify request failures, response latency, and backend bottlenecks under simulated concurrent-user loads.
- Evaluated Faster-Whisper transcription pipelines using audio chunking, overlapping windows, and Voice Activity Detection (VAD), and validated QR-based ticket-scanning workflows including attendee check-in and duplicate scan detection.

Technical Skills

Languages: Python, C++, Java, JavaScript
Backend & APIs: FastAPI, Node.js, Express.js, RESTful APIs, WebSockets
Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS
AI & Speech Processing: Gemini, ElevenLabs Scribe v2, Faster-Whisper, Speech-to-Text, Audio Chunking, VAD
Databases: PostgreSQL, MongoDB, MySQL
Testing & Monitoring: Apache JMeter, Grafana, API Testing, Performance Testing, Load Testing
Tools & DevOps: Docker, Docker Compose, Git, GitHub, Postman, n8n, AWS, GCP, Cloudflare Turnstile
Core Concepts: Data Structures & Algorithms, OOP, Asynchronous Programming

Projects

RoadResQ – Roadside Assistance Platform | *MERN Stack, Socket.IO, Leaflet, JWT* GitHub

- Built a real-time roadside assistance platform with live geolocation tracking using Leaflet and Socket.IO, enabling continuous location updates and efficient coordination between users and service providers.
- Integrated OpenStreetMap with Leaflet.js for real-time tracking and geolocation services
- Implemented secure JWT authentication with role-based dashboards and deployed on cloud platforms

MediQ – Hospital Management System | *React.js, Node.js, Express.js, MongoDB, JWT* GitHub

- Developed full-stack hospital management system with JWT authentication and role-based access control, managing patient workflows and resource allocation
- Designed REST APIs for OPD queue management, bed allocation, and inventory using Node.js and Express.js
- Designed and optimized MongoDB schema and deployed application using Vercel and Render with CI/CD pipeline

Real-Time Sales Streaming Pipeline | *Kafka, PostgreSQL, Python, Streamlit* GitHub

- Engineered a scalable end-to-end real-time data pipeline using **Kafka Producer–Consumer**, streaming continuous sales data into PostgreSQL
- Designed a **real-time dashboard** with Streamlit updating every 30 seconds to visualize revenue, trends, and live KPIs
- Implemented automated data generation using Faker and ensured seamless ingestion through Kafka topics into a structured data warehouse
- Structured modular architecture and documentation for scalability, deployment, and reproducibility

Certifications & Achievements

Oracle Certified Professional: Java SE 11 Developer | Oracle 2025

AWS Fundamentals Specialization | Amazon Web Services 2025

Google Cloud Platform – 15 Skill Badges | Cloud Computing & AI 2024

Research Paper Co-Author – IEEE CI2A 2026 | IEEE Xplore Publication 2026